

| Question |  |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|--|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (iii) |  | any three of following for (1) each<br><br><ul style="list-style-type: none"><li>potassium's outer shell electron furthest away from nucleus</li><li>least attraction / electron most easily lost</li><li>chlorine's outer shell closest to nucleus</li><li>greatest attraction / electron most easily gained</li></ul><br>allow (1) for 'reactivity increases down Group 1 and decreases down Group 7' if no other mark awarded | 3               |     |     | 3     |       |      |
|          |  |       |  | Question 6 total                                                                                                                                                                                                                                                                                                                                                                                                                 | 7               | 3   | 2   | 12    | 3     | 4    |